

CardioFusion

ECG model progress

1. The first model used separate segments

Each site recording was split into five-second segments and treated as a separate input.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.6858	63.8%	22.0%	64.2%	32.8%
Test	0.7921	65.8%	8.8%	77.5%	15.9%

A simple CNN with one input channel.

It caught some HFrEF cases, but produced too many false alarms.

Test confusion matrix

Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	1802	958
HFrEF	27	93

2. Changes for a four-channel model

Dataset review showed that each patient had four recordings that needed to be used together.

Group APEX, LLSB, LUSB and RUSB by patient ID.

Stack the four 30-second signals into one input.

Change the CNN from one input channel to four.

Add residual convolution blocks and split the data by patient.

3. Four-channel model results

One patient now gives one input: APEX, LLSB, LUSB and RUSB, each with 30 seconds.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.9564	91.0%	57.9%	91.7%	71.0%
Test	0.9983	96.7%	55.6%	100.0%	71.4%

The CNN now takes four channels together.
Deeper residual blocks learn richer features.

The model can combine information from
the same patient across the four sites.

Test confusion matrix

Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	111	4
HFrEF	0	5

4. Training with five-second segments

The four channels stay together. A fresh model learns from six windows per training patient.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.8845	87.0%	47.6%	83.3%	60.6%
Test	0.9200	83.3%	17.4%	80.0%	28.6%

These results use only the first five seconds.

The model still finds useful information,
but false positives rise from 4 to 19
compared with the full 30-second model.

Test confusion matrix

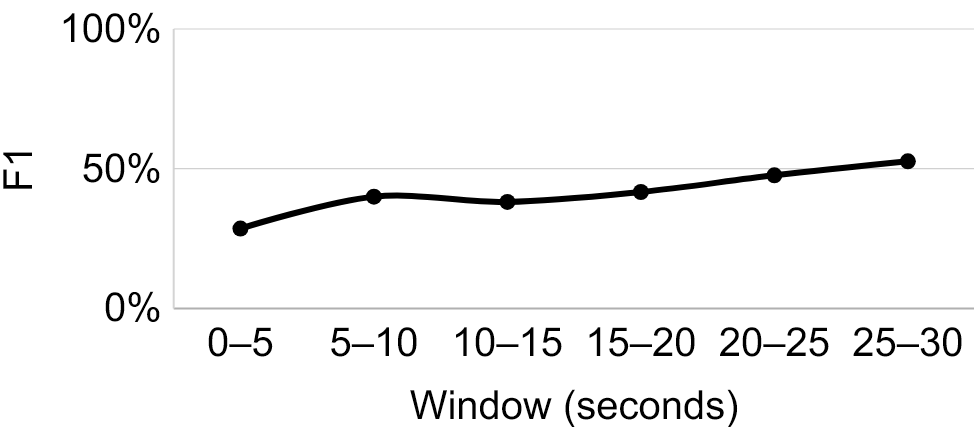
Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	96	19
HFrEF	1	4

5. Predictions change across segments

Test results for each five-second window, using the same model and cutoff.

All six predictions agree for 96 of 120 patients.

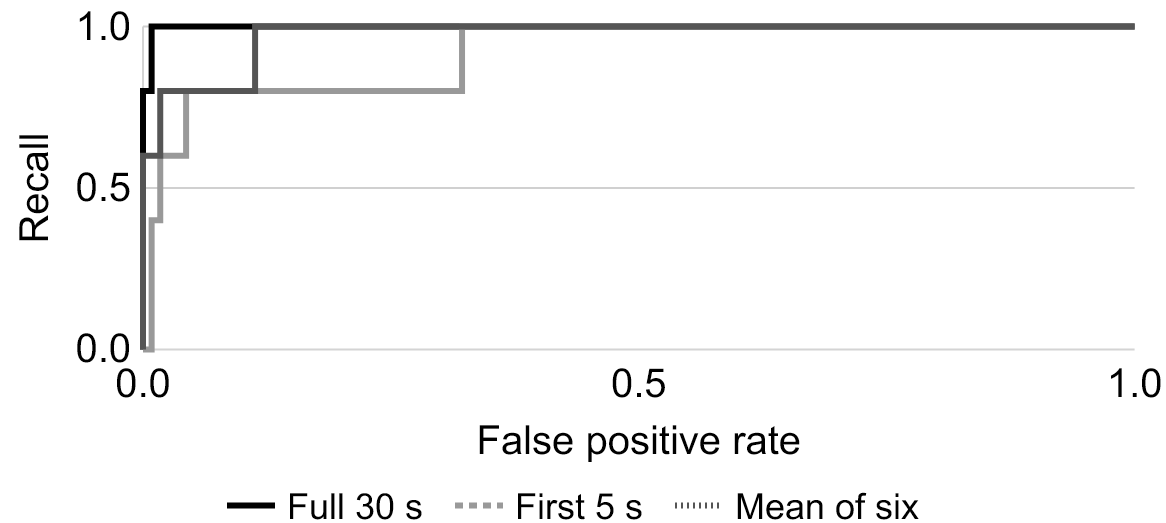
One HFrEF patient changes from negative in the first three windows to positive in the last three.



Results	AUROC	Accuracy	Precision	Recall	F1
0-5s	0.9200	83.3%	17.4%	80.0%	28.6%
5-10s	0.9357	90.0%	26.7%	80.0%	40.0%
10-15s	0.9670	89.2%	25.0%	80.0%	38.1%
15-20s	0.9635	88.3%	26.3%	100.0%	41.7%
20-25s	0.9739	90.8%	31.2%	100.0%	47.6%
25-30s	0.9896	92.5%	35.7%	100.0%	52.6%

6. Averaging segments helps

Comparison of one window, the mean of six window scores, and majority voting.



Averaging reduces false positives from 19 to 4.
It still misses one HFrEF case.

Results	AUROC	Accuracy	Precision	Recall	F1
First 5 seconds	0.9200	83.3%	17.4%	80.0%	28.6%
Mean of six (30 s)	0.9739	95.8%	50.0%	80.0%	61.5%
Majority vote (30 s)	0.9661	92.5%	33.3%	80.0%	47.1%
Full 30-second model	0.9983	96.7%	55.6%	100.0%	71.4%